

Total No. of Questions : 6]

SEAT No. :

P47

[Total No. of Pages : 2

APR - 18/TE/Insem. - 49

T.E. (Information Technology)

MULTIMEDIA TECHNOLOGIES

(2012 Course) (314452) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) State and explain various components of multimedia. Also, explain the need of multimedia. [5]

b) Write and explain Huffman Coding algorithm for the input with frequencies of occurrences A(3) B(5) 8(E) H(2) I(7). Construct Huffman Tree. [5]

OR

Q2) a) Define Multimedia. Why is multimedia important? Explain any two web applications. [5]

b) Explain Arithmetic coding Algorithm. Encode the message "a1a2a3a3a4" with initial probabilities of $a1 = 0.2$, $a2 = 0.2$, $a3 = 0.4$, $a4 = 0.2$. [5]

Q3) a) Explain any two smoothing and sharpening filters used for image enhancement. [6]

b) Write a short note on CMY and HSV color model. [4]

OR

Q4) a) Explain Vector Quantization lossy image compression. [5]

b) Explain JPEG image compression with neat sketch. [5]

P.T.O.

- Q5) a)** Write and explain MPEG Audio coding algorithm. **[5]**
- b)** What is MIDI? Explain MIDI with MIDI file structure and MIDI sound attribute. **[5]**

OR

- Q6) a)** Explain Predictive coding and delta modulation audio compression techniques. **[6]**
- b)** Explain WAV, WMA, MP3 Audio file formats. **[4]**

